



UNSW
SYDNEY

MATH2601 – Higher Linear Algebra

Topic 3 – Linear transformations

Lecture 3.1 – Linear transformations as maps

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Linear transformations

Definition. Suppose $(V, \mathbb{F})$ and $(W, \mathbb{F})$ are vector spaces over a common scalar field. A **linear transformation** (or **linear map** or **vector space homomorphism**) from V to W is a function $T : V \rightarrow W$ for which both

- $T(\mathbf{u} + \mathbf{v}) = T(\mathbf{u}) + T(\mathbf{v})$ for all $\mathbf{u}, \mathbf{v} \in V$, and
- $T(\lambda \mathbf{v}) = \lambda T(\mathbf{v})$ for all $\mathbf{v} \in V$ and $\lambda \in \mathbb{F}$.

Note this means that a linear transformation is a group homomorphism between the groups $(V, +)$ and $(W, +)$, with the additional property that it also preserves the structure imposed by scalar multiplication.

Example. The following are all linear transformations:

- The **identity map**, $\text{id} : V \rightarrow V$ for any vector space V .
- The **trivial map**, $T : V \rightarrow W$ given by $T(\mathbf{v}) = \mathbf{0}$ for all $\mathbf{v} \in V$.
- **Rotations**, **reflections**, and **projections** in $\mathbb{R}^n$ (see MATH1241).
- **Differentiation** and **definite integration** in $\mathcal{C}(\mathbb{R})$.

Lemma. For any linear transformation $T : V \rightarrow W$, we have $T(\mathbf{0}) = \mathbf{0}$ and $T(-\mathbf{v}) = -T(\mathbf{v})$ for all $\mathbf{v} \in V$.

Proof. These follow immediately from results we proved earlier about vector spaces: $T(\mathbf{0}) = T(0\mathbf{v}) = 0T(\mathbf{v}) = \mathbf{0}$, and $T(-\mathbf{v}) = T(-1\mathbf{v}) = -T(\mathbf{v})$.

Linearity test

Lemma. (Linearity test lemma)

Given two vector spaces $(V, \mathbb{F})$ and $(W, \mathbb{F})$, a function $T : V \rightarrow W$ is a linear transformation if and only if $T(\mathbf{u} + \lambda\mathbf{v}) = T(\mathbf{u}) + \lambda T(\mathbf{v})$ for all $\mathbf{u}, \mathbf{v} \in V$ and $\lambda \in \mathbb{F}$.

Proof. The forward direction is obvious. For the reverse direction, suppose $T(\mathbf{u} + \lambda\mathbf{v}) = T(\mathbf{u}) + \lambda T(\mathbf{v})$ for all $\mathbf{u}, \mathbf{v} \in V$ and $\lambda \in \mathbb{F}$.

Notice that $T(\mathbf{u} + \mathbf{v}) = T(\mathbf{u} + 1\mathbf{v}) = T(\mathbf{u}) + 1T(\mathbf{v}) = T(\mathbf{u}) + T(\mathbf{v})$ for any $\mathbf{u}, \mathbf{v} \in V$, showing T preserves vector addition.

Also, for any $\mathbf{v} \in V$ we have $T(\mathbf{0}) = T(\mathbf{v} + (-1)\mathbf{v}) = T(\mathbf{v}) + (-1)T(\mathbf{v}) = \mathbf{0}$.

So $T(\lambda\mathbf{v}) = T(\mathbf{0} + \lambda\mathbf{v}) = T(\mathbf{0}) + \lambda T(\mathbf{v}) = \mathbf{0} + \lambda T(\mathbf{v}) = \lambda T(\mathbf{v})$ for any $\mathbf{v} \in V$ and $\lambda \in \mathbb{F}$, showing T preserves scalar multiplication.

Thus T is linear.

Exercise. Let $A \in M_{p,q}(\mathbb{F})$. Show that $T : \mathbb{F}^q \rightarrow \mathbb{F}^p$ given by $T(\mathbf{v}) = A\mathbf{v}$ for all $\mathbf{v} \in \mathbb{F}^q$ is a linear transformation.

Solution. For any $\mathbf{u}, \mathbf{v} \in \mathbb{F}^q$ and $\lambda \in \mathbb{F}$ we have

$$T(\mathbf{u} + \lambda\mathbf{v}) = A(\mathbf{u} + \lambda\mathbf{v}) = A\mathbf{u} + \lambda A\mathbf{v} = T(\mathbf{u}) + \lambda T(\mathbf{v}),$$

so T is linear by the linearity test lemma.

Linear transformation examples

Exercise. Show that $T : \mathcal{P}(\mathbb{F}) \rightarrow \mathbb{F}^2$ given by $T(p) = \begin{pmatrix} p(1) \\ p(2) \end{pmatrix}$ for all $p \in \mathcal{P}(\mathbb{F})$ is linear.

Solution. Let $p, q \in \mathcal{P}(\mathbb{F})$ and $\lambda \in \mathbb{F}$. Then

$$T(p + \lambda q) = \begin{pmatrix} (p + \lambda q)(1) \\ (p + \lambda q)(2) \end{pmatrix} = \begin{pmatrix} p(1) + \lambda q(1) \\ p(2) + \lambda q(2) \end{pmatrix} = T(p) + \lambda T(q),$$

so T is linear by the linearity test.

Exercise. Given $(V, \mathbb{F})$ is a vector space with ordered basis $B = \{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n\}$, show that the coordinate function $T : V \rightarrow \mathbb{F}^n$ given by $T(\mathbf{u}) = [\mathbf{u}]_B$ for all $\mathbf{u} \in V$ is linear.

Solution. Let $\mathbf{a}, \mathbf{b} \in V$ and $\lambda \in \mathbb{F}$. Write $\mathbf{a} = a_1\mathbf{v}_1 + a_2\mathbf{v}_2 + \dots + a_n\mathbf{v}_n$ and $\mathbf{b} = b_1\mathbf{v}_1 + b_2\mathbf{v}_2 + \dots + b_n\mathbf{v}_n$ for coordinates $a_i, b_i \in \mathbb{F}$. Then

$$\mathbf{a} + \lambda\mathbf{b} = \sum_{i=1}^n a_i\mathbf{v}_i + \lambda \sum_{i=1}^n b_i\mathbf{v}_i = \sum_{i=1}^n (a_i + \lambda b_i)\mathbf{v}_i,$$

meaning $[\mathbf{a} + \lambda\mathbf{b}]_B = [\mathbf{a}]_B + \lambda[\mathbf{b}]_B$. So $T(\mathbf{a} + \lambda\mathbf{b}) = T(\mathbf{a}) + \lambda T(\mathbf{b})$, meaning T is linear by the linearity test.

Space of linear transformations

Definition. Given two vector spaces $(V, \mathbb{F})$ and $(W, \mathbb{F})$, the set of **all linear transformations** from V to W is denoted $\mathcal{L}(V, W)$.

Theorem. For any vector spaces $(V, \mathbb{F})$ and $(W, \mathbb{F})$, the set $\mathcal{L}(V, W)$ is a **vector space** (with the standard operations $(S + T)(\mathbf{v}) = S(\mathbf{v}) + T(\mathbf{v})$ for all $S, T \in \mathcal{L}(V, W)$ and $(\lambda T)(\mathbf{v}) = \lambda(T(\mathbf{v}))$ for all $T \in \mathcal{L}(V, W)$ and $\lambda \in \mathbb{F}$).

Proof. First note that $\mathcal{L}(V, W)$ is nonempty, since $Z \in \mathcal{L}(V, W)$ where $Z : V \rightarrow W$ is the trivial linear map given by $Z(\mathbf{v}) = \mathbf{0}$ for all $\mathbf{v} \in V$. Now let $S, T \in \mathcal{L}(V, W)$ and $\lambda \in \mathbb{F}$. Per the subspace lemma (with respect to a generalisation of $\mathcal{F}(V)$), we want to show $S + \lambda T \in \mathcal{L}(V, W)$, that is we want to show $S + \lambda T$ is linear.

Let $\mathbf{u}, \mathbf{v} \in V$ and $\mu \in \mathbb{F}$. Then

$$\begin{aligned}(S + \lambda T)(\mathbf{u} + \mu\mathbf{v}) &= S(\mathbf{u} + \mu\mathbf{v}) + \lambda T(\mathbf{u} + \mu\mathbf{v}) \\ &= S(\mathbf{u}) + \mu S(\mathbf{v}) + \lambda T(\mathbf{u}) + \mu(\lambda T(\mathbf{v})) \quad (S, T \text{ linear}) \\ &= (S + \lambda T)(\mathbf{u}) + \mu(S + \lambda T)(\mathbf{v}),\end{aligned}$$

so $S + \lambda T$ is indeed linear by the linearity test.

Corollary. The sum of two linear maps is also linear, as is any scalar multiple of a linear map.

Other derived transformations

Lemma. If $S : U \rightarrow V$ and $T : V \rightarrow W$ are linear transformations, then their composition $T \circ S : U \rightarrow W$ is also linear.

Proof. For all $\mathbf{u}, \mathbf{v} \in U$ and $\lambda \in \mathbb{F}$ (for underlying scalar field $\mathbb{F}$), we have $T \circ S(\mathbf{u} + \lambda \mathbf{v}) = T(S(\mathbf{u}) + \lambda S(\mathbf{v})) = T(S(\mathbf{u})) + \lambda(T(S(\mathbf{v})))$, so $T \circ S(\mathbf{u} + \lambda \mathbf{v}) = T \circ S(\mathbf{u}) + \lambda(T \circ S(\mathbf{v}))$ meaning $T \circ S$ is linear.

Lemma. If $T : V \rightarrow W$ is a linear transformation that is invertible, then $T^{-1} : W \rightarrow V$ is also linear.

Proof. Let $\mathbf{u}, \mathbf{v} \in W$ and $\lambda \in \mathbb{F}$ (for underlying scalar field $\mathbb{F}$). Then we may write $\mathbf{a} = T^{-1}(\mathbf{u})$ and $\mathbf{b} = T^{-1}(\mathbf{v})$ for some unique $\mathbf{a}, \mathbf{b} \in V$. Then $T^{-1}(\mathbf{u} + \lambda \mathbf{v}) = T^{-1}(T(\mathbf{a}) + \lambda T(\mathbf{b})) = T^{-1}(T(\mathbf{a} + \lambda \mathbf{b})) = \mathbf{a} + \lambda \mathbf{b}$, so $T^{-1}(\mathbf{u} + \lambda \mathbf{v}) = T^{-1}(\mathbf{u}) + \lambda T^{-1}(\mathbf{v})$ meaning T^{-1} is linear.

Lemma. Let $G = \{T \in \mathcal{L}(V, V) : T \text{ is invertible}\}$. Then G is a group under composition.

Proof. Since compositions of linear maps are linear, G is closed under composition. Associativity is an inherent property of function composition. The identity in G is the identity function, since $\text{id} \circ T = T \circ \text{id} = T$ for all $T \in G$. Since inverses of invertible linear maps are linear, G is closed under taking inverses. So $(G, \circ)$ is a group.

Kernel and image

We can define the kernel and image of vector space homomorphisms in exactly the same way they were defined earlier for group homomorphisms.

Definition. The **kernel** of a linear transformation $T : V \rightarrow W$ is the set

$$\ker(T) = \{\mathbf{v} \in V : T(\mathbf{v}) = \mathbf{0}\}.$$

Definition. The **image** of a linear transformation $T : V \rightarrow W$ is the set

$$\operatorname{im}(T) = \{T(\mathbf{v}) : \mathbf{v} \in V\}.$$

Example. Recall $T : \mathcal{P}(\mathbb{R}) \rightarrow \mathbb{R}^2$ given by $T(p) = \begin{pmatrix} p(0) \\ p(1) \end{pmatrix}$ is linear.

In this case, $\operatorname{im}(T) = \mathbb{R}^2$, while $\ker(T) = \{t(t-1)p(t) : p \in \mathcal{P}(\mathbb{R})\}$.

It is also convenient to define the image of a subset of V under T .

Definition. Given a linear transformation $T : V \rightarrow W$, the image of a subset $U \subseteq V$ is the set $T(U) = \{T(\mathbf{u}) : \mathbf{u} \in U\}$.

Using this notation, for a linear transformation $T : V \rightarrow W$ we have $\operatorname{im}(T) = T(V)$.

Kernel and image subspaces

Lemma. Given a linear transformation $T : V \rightarrow W$, if $U \leq V$ then $T(U) \leq W$.

Proof. Clearly $T(U)$ is nonempty since U is (since it's a vector space). Let $\mathbf{u}, \mathbf{v} \in T(U)$ and $\lambda \in \mathbb{F}$ (for underlying scalar field $\mathbb{F}$). Then there are some $\mathbf{a}, \mathbf{b} \in U$ such that $T(\mathbf{a}) = \mathbf{u}$ and $T(\mathbf{b}) = \mathbf{v}$. Notice that $\mathbf{u} + \lambda\mathbf{v} = T(\mathbf{a}) + \lambda T(\mathbf{b}) = T(\mathbf{a} + \lambda\mathbf{b})$ by the linearity of T . Since U is a subspace, $\mathbf{a} + \lambda\mathbf{b} \in U$ and so $T(\mathbf{u} + \lambda\mathbf{v}) \in T(U)$. So $T(U) \leq W$ by the subspace lemma.

Theorem. Given a linear transformation $T : V \rightarrow W$, we have $\ker(T) \leq V$ and $\text{im}(T) \leq W$ (as subspaces).

Proof. From the above lemma, we immediately get $\text{im}(T) = T(V) \leq W$.

For the kernel, first note $\ker(T)$ is nonempty since certainly $\mathbf{0} \in \ker(T)$. Let $\mathbf{u}, \mathbf{v} \in \ker(T)$ and $\lambda \in \mathbb{F}$ (for underlying scalar field $\mathbb{F}$). Then $T(\mathbf{u}) = T(\mathbf{v}) = \mathbf{0}$, so $T(\mathbf{u} + \lambda\mathbf{v}) = T(\mathbf{u}) + \lambda T(\mathbf{v}) = \mathbf{0}$, meaning $\mathbf{u} + \lambda\mathbf{v} \in \ker(T)$ and so $\ker(T) \leq V$ by the subspace lemma.

Rank and nullity

Definition. Given a linear transformation $T : V \rightarrow W$, the **nullity** of T is the dimension of its kernel, written $\text{nullity}(T) = \dim(\ker(T))$, and the **rank** of T is the dimension of its image, written $\text{rank}(T) = \dim(\text{im}(T))$.

Example. For any $n \in \mathbb{Z}^+$, let $D : \mathcal{P}_n(\mathbb{R}) \rightarrow \mathcal{P}_n(\mathbb{R})$ be the linear transformation given by differentiation (that is, $D(p(t)) = \frac{d}{dt}p(t)$). Then $\text{nullity}(D) = 1$ since $\ker(D) = \mathcal{P}_0(\mathbb{R})$, and $\text{rank}(D) = n$ since $\text{im}(D) = \mathcal{P}_{n-1}(\mathbb{R})$.

Lemma. A linear transformation $T : V \rightarrow W$ is **injective** if and only if $\text{nullity}(T) = 0$.

Proof. For the forward direction, since $T(\mathbf{0}) = \mathbf{0}$, if T is injective then we must have $\ker(T) = \{\mathbf{0}\}$, whence $\text{nullity}(T) = \dim(\ker(T)) = 0$.

For the reverse direction, if $\text{nullity}(T) = 0$ then $\ker(T) = \{\mathbf{0}\}$. For $\mathbf{u}, \mathbf{v} \in V$, if $T(\mathbf{u}) = T(\mathbf{v})$ then $T(\mathbf{u} - \mathbf{v}) = T(\mathbf{u}) - T(\mathbf{v}) = \mathbf{0}$, so since the kernel is trivial we must have $\mathbf{u} - \mathbf{v} = \mathbf{0}$, meaning $\mathbf{u} = \mathbf{v}$ and so T is injective.

Rank-nullity theorem

Theorem. (Rank-nullity theorem)

For any linear transformation $T : V \rightarrow W$, $\text{rank}(T) + \text{nullity}(T) = \dim(V)$.
That is, $\dim(V) = \dim(\ker(T)) + \dim(\text{im}(T))$.

Proof. If $\text{rank}(T) = \infty$ or $\text{nullity}(T) = \infty$, the result is trivially true, so suppose $\ker(T)$ is finite-dimensional and let $\text{nullity}(T) = n$ for some $n \in \mathbb{N}$. Further let $A = \{\mathbf{a}_1, \dots, \mathbf{a}_n\}$ be a basis for $\ker(T)$. We know that we can extend this to a basis for V , say $B = \{\mathbf{a}_1, \dots, \mathbf{a}_n, \mathbf{b}_1, \dots, \mathbf{b}_r\}$ for some $r \in \mathbb{N}$. Then we wish to show $\text{rank}(T) = r$.

Consider the set $C = \{T(\mathbf{b}_1), \dots, T(\mathbf{b}_r)\}$. Notice that for any $\mathbf{v} \in V$, we have $\mathbf{v} = \alpha_1 \mathbf{a}_1 + \dots + \alpha_n \mathbf{a}_n + \beta_1 \mathbf{b}_1 + \dots + \beta_r \mathbf{b}_r$ for some $\alpha_i, \beta_i \in \mathbb{F}$, the underlying scalar field. Then

$$T(\mathbf{v}) = T(\sum \alpha_i \mathbf{a}_i + \sum \beta_i \mathbf{b}_i) = \sum \alpha_i T(\mathbf{a}_i) + \sum \beta_i T(\mathbf{b}_i) = \sum \beta_i T(\mathbf{b}_i),$$

since each $\mathbf{a}_i \in \ker(T)$. So $T(V) = \text{im}(T)$ is spanned by C .

Now suppose $\beta_1 T(\mathbf{b}_1) + \dots + \beta_r T(\mathbf{b}_r) = \mathbf{0}$ for some $\beta_i \in \mathbb{F}$. Then $T(\sum \beta_i \mathbf{b}_i) = \mathbf{0}$, so $\sum \beta_i \mathbf{b}_i \in \ker(T) = \text{span}(A)$. So the unique coordinates for $\sum \beta_i \mathbf{b}_i$ over B in V must have all $\mathbf{b}_i$ coefficients equal to 0. That is, each $\beta_i = 0$, meaning C is linearly independent.

So C is a basis for $\text{im}(T)$ and hence $\text{rank}(T) = r$ as required.

Bijjective linear maps

Theorem. Given a linear transformation $T : V \rightarrow W$ for finite-dimensional vector spaces V and W with $\dim(V) = \dim(W)$, if T is injective or T is surjective, then it is both.

Proof. If T is injective, then $\text{nullity}(T) = 0$. Then by the rank-nullity theorem, $\text{rank}(T) = \dim(V) = \dim(W)$. Since $\text{im}(T) \leq W$ and they have the same dimension, we must have $\text{im}(T) = W$, meaning T is surjective.

Similarly, if T is surjective then $\text{rank}(T) = \dim(W) = \dim(V)$, so by the rank-nullity theorem $\text{nullity}(T) = 0$, which means T is injective.

Corollary. Given a linear transformation $T : V \rightarrow W$ for finite-dimensional vector spaces V and W with $\dim(V) = \dim(W)$, the function T is bijective if and only if $\text{nullity}(T) = 0$ or $\text{rank}(T) = \dim(V)$.

Exercise. Show $T : \mathcal{P}_2(\mathbb{R}) \rightarrow \mathbb{R}^3$ given by $T(p) = \begin{pmatrix} p(1) \\ p(2) \\ p(3) \end{pmatrix}$ is bijective.

Solution. It is easy to check that T is linear. Since $\dim(\mathcal{P}_2(\mathbb{R})) = \dim(\mathbb{R}^3)$, we only have to check that T is either injective or surjective. Notice that if $p \in \ker(T)$ then $p(1) = p(2) = p(3) = 0$. Since $p \in \mathcal{P}_2(\mathbb{R})$, the only way a polynomial of degree at most 2 can have 3 distinct roots is if it is the zero polynomial. So $\text{nullity}(T) = 0$, meaning T is injective and thus bijective.

Vector space isomorphisms

Definition. A **vector space isomorphism** is a **bijective** linear transformation. If there is an isomorphism between two vector spaces V and W , we say V and W are **isomorphic** and write $V \cong W$ (thought of as “the same” space).

Theorem. Two finite-dimensional vector spaces $(V, \mathbb{F})$ and $(W, \mathbb{F})$ are **isomorphic** if and only if $\dim(V) = \dim(W)$.

Proof. For the forward direction, given an isomorphism $T : V \rightarrow W$, consider a basis $B = \{\mathbf{v}_1, \dots, \mathbf{v}_n\}$ for V . For any $\mathbf{w} \in W$, there is a corresponding $\mathbf{v} \in V$ for which $\mathbf{w} = T(\mathbf{v}) = T(\sum \lambda_i \mathbf{v}_i) = \sum \lambda_i T(\mathbf{v}_i)$, so $T(B)$ spans W . Furthermore, since $\mathbf{w} = \mathbf{0}$ implies $\mathbf{v} = \mathbf{0}$ (since $\ker(T)$ is trivial) which has unique basis coordinates $\lambda_i = 0$ for all i , we know $T(B)$ is linearly independent. Thus $T(B)$ is a basis for W , so $\dim(V) = \dim(W)$.

For the reverse direction, suppose $\dim(V) = \dim(W) = n$ for some $n \in \mathbb{N}$. We showed earlier that the coordinate map $T : V \rightarrow \mathbb{F}^n$ given by $T(\mathbf{v}) = [\mathbf{v}]_B$ for some basis B of V is a linear transformation. Since B is linearly independent, $\mathbf{0}$ is the unique solution to $T(\mathbf{v}) = \mathbf{0}$, so $\ker(T) = \{\mathbf{0}\}$ and T is bijective. So $V \cong \mathbb{F}^n$, and likewise $W \cong \mathbb{F}^n$, so $V \cong W$ since isomorphism is an equivalence relation.

This justifies why any vector space can be thought of as just its coordinates over $\mathbb{F}^n$ – because every vector space of dimension n is isomorphic to $\mathbb{F}^n$.